

Building Sector Intervention Details

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Intervention detail (in the calculator file)	Intervention (short) (in the calculator file)	Intervention description
What If there is no MetCouncil Plan: 2040 has the same single family (SF) homes proportion as 2018 (single family homes are larger than Multi family, and typically use more energy)	Met Council MF plan effects	Met Council's projection to 2040 already assumed a significant compact development. To evaluate the carbon mitigation potential, we first assume the % of single-family homes in 2040 is equivalent to 2018' and then calculate the energy reduction due to more multi-family homes in the default baseline projection.
Compact buildings: half of new SF homes become (multi family (MF)	Compact buildings (50% MF new)	In the compact development scenario (beyond Met Council's projection), we assume 50% (the default value) of the new single-family homes to be multi-family homes. Cities can adjust the defaulted percentage according to local preference.
Apply 0.8% reduction in Electricity EUI (Energy Use Intensity) for 1% tree canopy change, scale the impact accordingly.	Tree canopy energy reduction	A 1% increase in tree canopy coverage can reduce electricity use intensity by 0.8%. This elasticity is extracted from analysis using data in the city of St. Paul.
Affordable Floor Area (FA); All SF floor area grows only by 5% vs now at 15%	Affordable floor area	Cities can control the size of new single-family homes to reduce energy use. The growth of the single-family home size is reduced from 15% to 5% to make the new single-family home smaller. We apply this impact to 50% of new single-family homes. Cities can adjust the growth rate of single-family homes' floor area and the percentage of new single-family homes being controlled.
New home efficiency; 50% new homes to LEED Gold (25 kBtu/sf; electricity reduces by 64% and gas by 64%)	New homes (50%) are LEED-Gold	Cities can adopt strict building codes requiring new homes to achieve LEED-Gold standards. We assumed 50% of new homes (single- and multi-family homes) to be LEED-Gold with both electricity and gas EUI reduced 64%. Cities can adjust the percentage of new homes achieving this standard.
Existing home: 80% reduced EUI by 33%	Existing homes (80%) retrofitted	Cities can retrofit existing homes to be energy efficient. We assume 80% of existing homes with energy use intensity reduced by 33%. Cities can adjust the percentage of existing homes to be retrofitted.

Passive Buildings for remaining 20% stock at 66% reduction	Existing homes (20%), passive homes	Cities can target large existing homes to be more efficient. We assume 20% of floor areas (that are the largest homes) reducing EUI by 66%, which is close to passive homes' energy use standard. Cities can adjust the percentage of the largest homes to be passive homes.
80% of existing commercial buildings are LEED Gold (reduce EUI by 25%)	Existing commercial bldgs (80%), LEED-Gold	Cities can retrofit existing commercial buildings to make them more efficient. The default is 80% of existing commercial buildings will be retrofitted to reduce EUI by 25%. Cities can adjust the percentage of commercial buildings to be retrofitted.
All homes receive effective messages, use in-home display, and smart meters (reduce HH energy use by 11%)	Behavioral & smart nudges (100% HH)	Efficient messaging, in-home energy display, and smart meters together can reduce household energy use by 11%. The default is 100% of homes change behaviors due to these actions. Cities can adjust this participation rate accordingly.
Smart grid and technologies reduce commercial electricity use by 12%	Smart grid efficiency (100% comm buldgs)	Smart grid and technologies by controlling energy use more efficiently can reduce commercial buildings energy use by 12%. The default is that the grid has been upgraded. Therefore, all commercial buildings will reduce energy use. The coverage of the impact (100% of commercial buildings) is fixed by grid.
Smart grid and technologies reduce industrial electricity use by 12%	Smart grid efficiency (100% ind)	Similar to commercial buildings, smart grid and technologies help to reduce industrial energy use by 12%.
Grid decarbonization: GHGs reduced 80% (compared to 2018)	Grid decarbonization, 113 kgCO ₂ e/MWh	Due to decarbonizing the grid, the emission factor of electricity decreases from 568 kg CO ₂ e/MWh (2018 Minnesota state level) to 113 kgCO ₂ e/MWh. This change is fixed by the grid.
59% of homes use ground source heat pumps and tankless water heaters	Electrifying heating (59% HH)	59% of homes are using air-source heat pumps and electric water heaters, instead of using gas for heating. Cities can adjust the percentage of homes participated in electrifying heating.
31.7% of natural gas replaced by renewable natural gas	RNG substitutes 31.7% natural gas in all sectors	About 31.7% of natural gas supply at the state level can be replaced by renewable natural gas from the state of Minnesota. This percentage is fixed by the gas infrastructure supply.